E-NEWS EXPRESS CASE STUDY

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CONTENTS / AGENDA

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- Business Problem Overview and Solution Approach
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- Hypotheses Tests and Results
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BUSINESS PROBLEM OVERVIEW CONTEXT

- E-news Express is an online new portal that want to increase their subscriber growth as they have noticed that they are having a decline in monthly subscribers.
- Since the current landing page may not be engaging, a new landing page was designed to improve user experience.
    - However, we need to determine if the new landing page improves the:
        - User engagement – time spent on each page
        - Conversion rate – subscribers
        - User behavior based on language preferences
- This directly impacts revenue, growth, and strategy.

EXECUTIVE SUMMARY - RECOMMENDATIONS

- E-news should make sure that their landing page is optimized over just looking at languages to ensure users stay on the page longer and convert to subscribers.
- E-news should be optimized to load quickly, have short and easy to read paragraphs, key points, and good quality images.
- E-news should also ensure that their landing page loads well on smartphones.
- E-news should continue to offer the provided language options in English, Spanish, and French, but also add local content to increase relevance based on location.

EXECUTIVE SUMMARY - RECOMMENDATIONS

- E-news can also move towards behavior based segmentation:
  - New subscribers vs. returning users
  - Users who click more vs those who don’t
  - Device type (smartphone vs Desktop)
- E-news can add customization and dynamic content blocks for improved conversion efficiency and less friction. This can also lead to higher ret

EXECUTIVE SUMMARY - CONCLUSION

- The E-news users that visited the new landing page spent more time on it than the old landing page. This means that the **new landing page successfully encouraged the users to stay on their website**.
    - P-value = 0.0003 < Significance level of 0.05. We reject the null hypothesis.
- The **users that were visiting the new landing page were more likely to convert** than the users that were visiting the old landing page.
    - P-value = 0.008 < Significance level of 0.05. We reject the null hypothesis.
- Based on the sample data, the number **of users that convert to subscribers is not dependent on the language preferred.**
    - P-value = 0.212 > Significance level of 0.05. We fail to reject the null hypothesis.
- The mean **time spent on the new landing page for all three languages are all equal.**
    - P-value = 0.43 > Significance level of 0.05. We fail to reject the null hypothesis.

BUSINESS MAIN PROBLEM

- Does the new landing page perform better than the old landing page for user engagement and subscriber conversions?
- We are testing if:
  - Do users spend more time on the new landing page now rather than the old one?
  - Does the new landing page lead to a higher conversion rate in subscribers?
  - Is the subscriber conversion also based on the language (English/French/Spanish)?
  - Is the time spent on the new page different based on different languages?

BUSINESS PROBLEM SOLUTION METHODOLOGY

1. Data Loading and Setup
2. Data Cleaning and Validation
3. Exploratory Data Analysis
4. Univariate Analysis
5. Bivariate Analysis
6. Defining Hypothesis Tests For Each Business Question
7. Selecting the Statistical Tests For Each Data Type
8. Setting the Significance Level
9. Compute and Compare the p-values with alpha
10. Draw Conclusions

EDA RESULTS – DATA OVERVIEW

The data contains information regarding the interaction of users in both groups with the two versions of the landing page.

- user_id - Unique user ID of the person visiting the website
- group - Whether the user belongs to the first group (control) or the second group (treatment)
- landing_page - Whether the landing page is new or old
- time_spent_on_the_page - Time (in minutes) spent by the user on the landing page
- converted - Whether the user gets converted to a subscriber of the news portal or not
- language_preferred - Language chosen by the user to view the landing page

There are 100 rows and 6 columns. There are no missing values or duplicates.

EDA RESULTS – DATA OVERVIEW

- Average time spent by users on a page = 5.38 mins
- Minimum time spent = 0.19 mins
- Max time spent = 10.71 mins
- 75% of people spent 7.02 mins on a page
- Time is also a continuous variable.

	user_id	time_spent_on_the_page
count	100.000000	100.000000
mean	546517.000000	5.377800
std	52.295779	2.378166
min	546443.000000	0.190000
25%	546467.750000	3.880000
50%	546492.500000	5.415000
75%	546567.250000	7.022500
max	546592.000000	10.710000

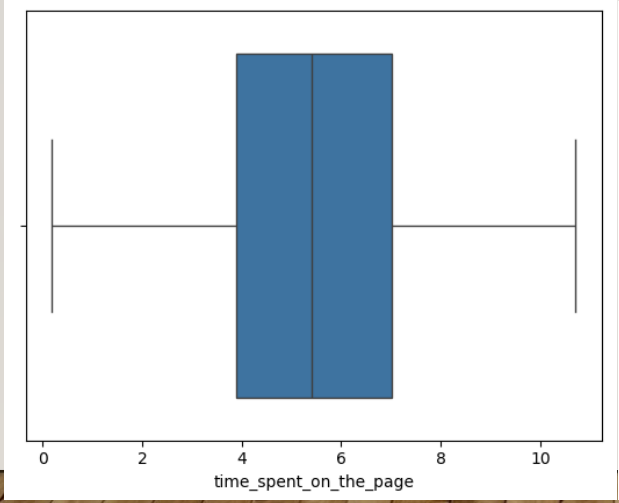
EDA RESULTS – DATA OVERVIEW

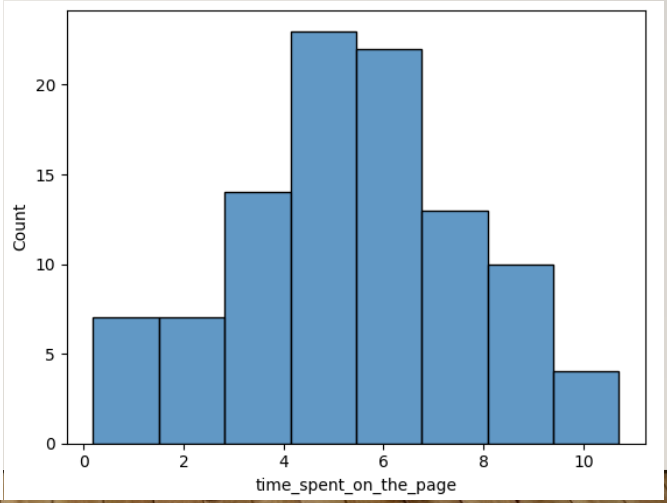
- The group is split into a control and new (treatment) group. Both have 50 entries each.
- The landing page has “old” and “new” values, representing the old landing page and the new landing page.
- Converted will represent “yes” if they converted or “no” if not.
- Language preferred consists of English, French, and Spanish. French/Spanish is the most preferred.

	count	unique	top	freq
group	100	2	control	50
landing_page	100	2	new	50
converted	100	2	yes	54
language_preferred	100	3	French	34

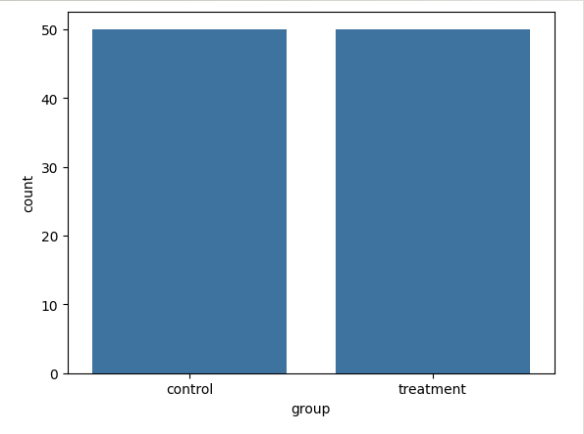
EDA RESULTS – UNIVARIATE ANALYSIS

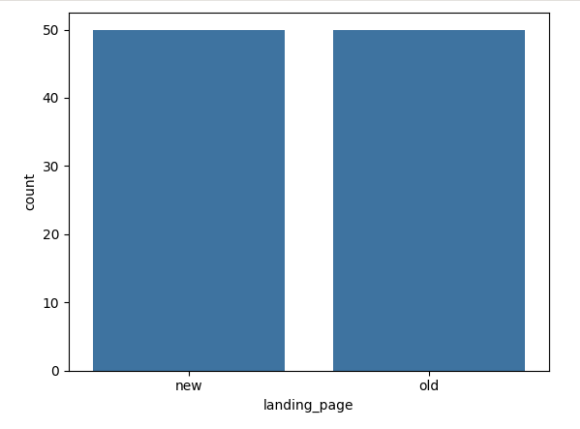
- The minimum time spent on the landing page is 0.19 minutes and the maximum time spent on the la nding page is 10.71 minutes.
- Time spent on the page is normally distributed across its mean of 5.37 minutes.

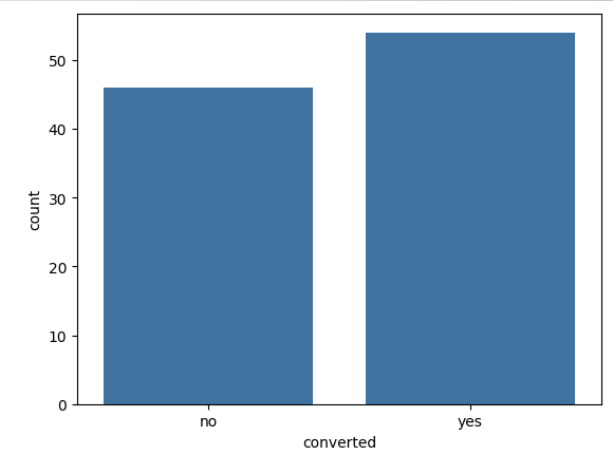




EDA RESULTS – UNIVARIATE ANALYSIS



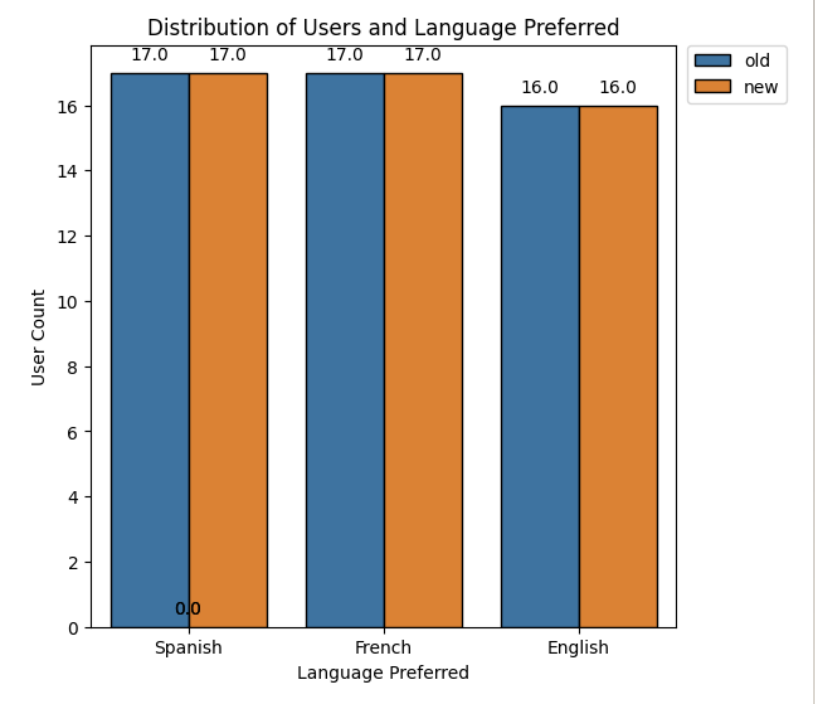




- The dataset is well-balanced with an equal number of users exposed to the new and old landing pages (50 each), ensuring a fair A/B test design. Overall, 54 users converted, indicating that there was a slight conversion rate across the sample.

EDA RESULTS – UNIVARIATE ANALYSIS

Language Preferred	
French	34
Spanish	34
English	32

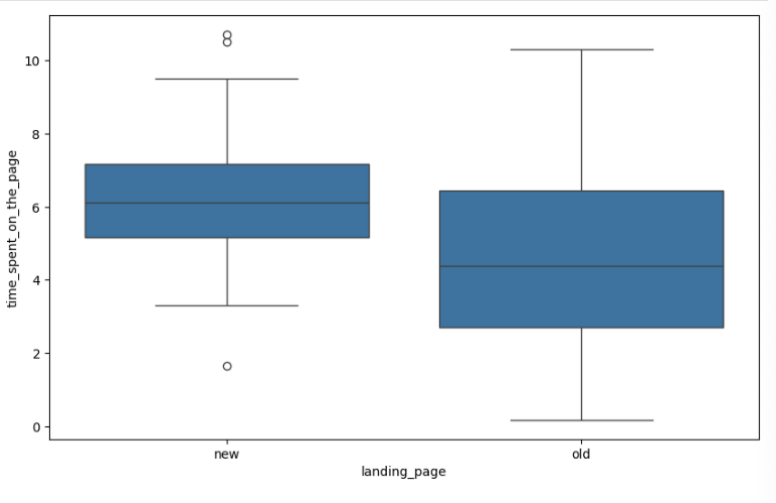


EDA RESULTS – UNIVARIATE ANALYSIS

Language Preferred	Converted	Percentage
English	Yes	65.625%
	No	34.375%
French	Yes	44.11%
	No	55.88%
Spanish	Yes	52.941%
	No	47.05%

- There are 3 different languages (English, French, and Spanish) that are used by users to read the old and new landing pages.
- About 65% of visitors who read the landing page in English became subscribers, followed by Spanish (52%), and by French (44%).

EDA RESULTS – BIVARIATE ANALYSIS



- New landing page: Median = 6, meaning that users spend more time on the new page
- Old landing page: Median = 4.5, meaning that users spend less time on the old page

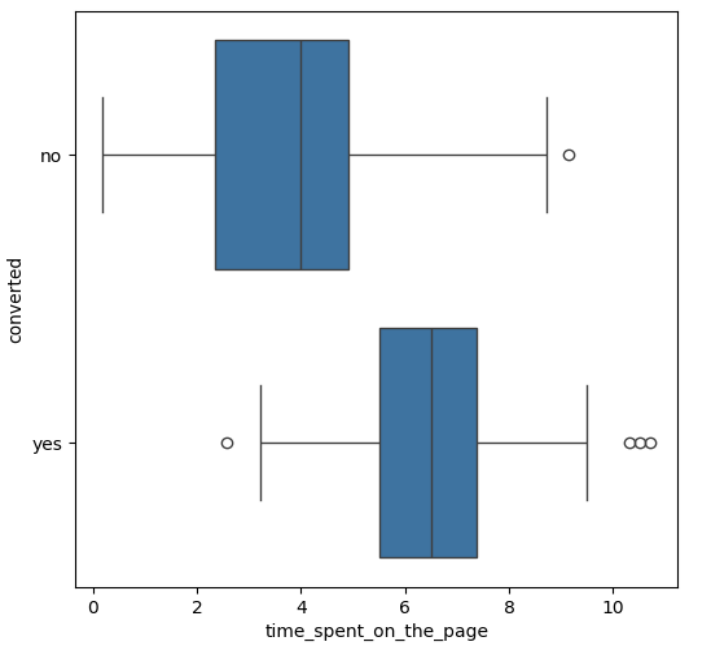
EDA RESULTS – BIVARIATE ANALYSIS

Users who DID NOT convert (no):

- Median = 3 mins, meaning that users w ho did not convert spent less time

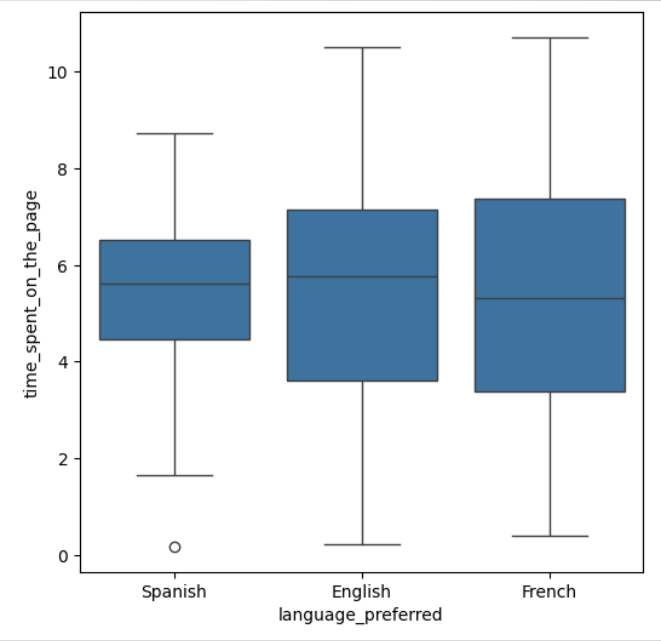
Users who DID convert (yes):

- Median = 6.0, meaning users who conv erted spent more time



EDA RESULTS – BIVARIATE ANALYSIS

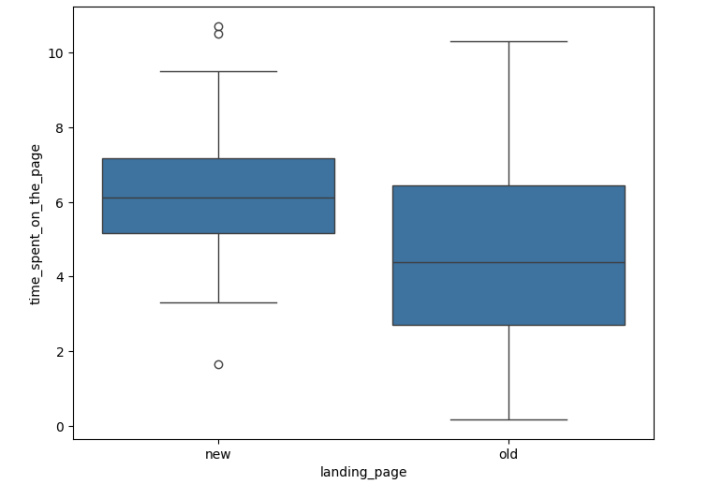
- All three language groups had a similar median time of 5-6 mins.
- This means that language preference has little effect on average engagement.
- The Spanish language is the most consistent and French speakers are the most variable.



EDA RESULTS – BIVARIATE ANALYSIS

Do the users spend more time on the new landing page than the existing landing page?

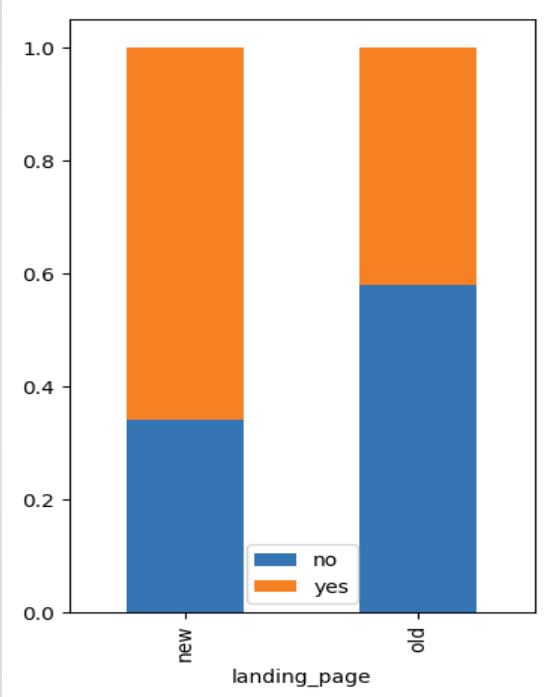
- Users spent more time on the new landing page.
  - Median of new page 6.1 > Median of old page 4.5
- Test: T-Test
  - We are comparing 2 means from independent grou ps with unknown and unequal standard deviations.
  - P-value = 0.00028 < 0.05 Significance level
  - We reject the null hypothesis
- Users spend more time on the new landing page.



EDA RESULTS – BIVARIATE ANALYSIS

Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?

- Users show a higher conversion on the new landing page.
- Test: One Tailed Z-test
  - We are comparing 2 proportions from independent groups.
  - P-value = 0.008 < 0.05 Significance level
  - We reject the null hypothesis

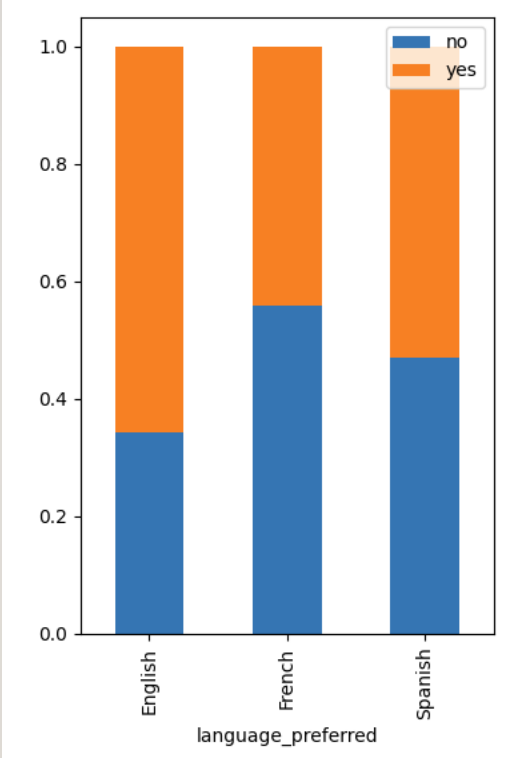


- The new landing page shows a significantly higher conversation rate to subscribers.

EDA RESULTS – BIVARIATE ANALYSIS

Does the converted status depend on the preferred language?

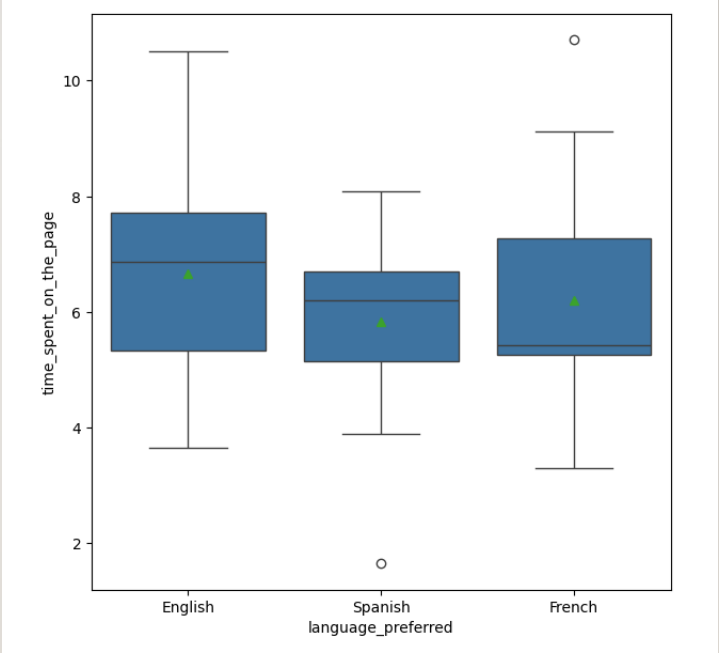
- English have a higher conversion rate. Spanish has a moderate conversion rate. French has the lowest conversion rate.
- Test: Chi-Square Test of Independence
- We are comparing 2 categorical variables across multiple categ ories simultaneously.
    - P-value = 0.213 > 0.05 Significance level
    - We fail to reject the null hypothesis.
- There is no strong evidence that conversion depends heavily on preferred language.



EDA RESULTS – BIVARIATE ANALYSIS

Is Time Spent on the New Page the Same Across Languages?

- Medians are similar across Spanish, English, and French on the new page
- Test: ANOVA
- We are testing all 3 groups whole controlling for false positives.
  - P-value = 0.432 > 0.05 Significance level
  - We fail to reject the null hypothesis.



- Time spent on the new page is not significantly different across lan guage groups

OVERALL SUMMARY

Question	Test Used	Why	P-value	Outcome
New page vs Old page time	T-test	2 means and unequal variances	0.00028	Users spend more time on new page.
Conversion rate	Z-test	2 proportions	0.008	New page has higher conversion rate.
Language vs Conversion	Chi-Square	2 categorical variables	0.213	No dependency on language vs. conversion
Languages vs Time	ANOVA	3 means	0.432	There's no difference.

APPENDIX

DATA BACKGROUND AND CONTENTS

- This is the first few rows of the data set.

	user_id	group	landing_page	time_spent_on_the_page	converted	language_preferred
0	546592	control	old	3.48	no	Spanish
1	546468	treatment	new	7.13	yes	English
2	546462	treatment	new	4.40	no	Spanish
3	546567	control	old	3.02	no	French
4	546459	treatment	new	4.75	yes	Spanish

DATA TYPES

- User_id is a integer value.
- time_spent_on_the_page is a float value.
- group, landing_page, converted, language_preferred is a string value.

```
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   user_id                 100 non-null    int64
 1   group                   100 non-null    str
 2   landing_page            100 non-null    str
 3   time_spent_on_the_page  100 non-null    float64
 4   converted               100 non-null    str
 5   language_preferred      100 non-null    str
dtypes: float64(1), int64(1), str(4)
memory usage: 4.8 KB
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